PROTEIN REQUIREMENTS GUIDE

Comprehensive Macro Planning Reference for Clinical & General Nutrition

1. Role of Protein in the Body

Function	Mechanism	Clinical Relevance
Muscle synthesis	mTOR pathway activation; leucine-triggered MPS (muscle protein synthesis)	Critical in sarcopenia, post-surgery recovery, sports nutrition
Metabolic regulation	Protein has highest TEF (thermic effect of food) — 20–30% of calories burned in digestion	Supports weight management and metabolic rate
Satiety & appetite control	Stimulates CCK, GLP-1, PYY; suppresses ghrelin	Reduces overall caloric intake; key in weight loss diets
Hormone & enzyme production	Amino acids are precursors to insulin, thyroid hormones, neurotransmitters, digestive enzymes	Protein deficiency impairs hormonal function
Immune function	Antibodies are glycoproteins; lymphocyte production requires amino acids	Low protein = impaired immune response
Wound healing & tissue repair	Collagen synthesis (requires glycine, proline, hydroxyproline) and cell proliferation	Post-surgery and injury nutrition
Transport proteins	Albumin transports fatty acids, drugs; hemoglobin transports O2	Hypoalbuminemia indicates malnutrition

2. Protein Recommendations by Goal & Population

Population / Goal	Protein Intake	Total for 70kg Person	Source
Sedentary adult (minimum)	0.8 g/kg/day	56 g/day	WHO/RDA
Generally active adult	1.0–1.2 g/kg/day	70–84 g/day	ICMR 2020
Weight loss (preserve muscle)	1.2–1.6 g/kg/day	84–112 g/day	ISSN, Layman et al.
Muscle gain (resistance training)	1.6–2.2 g/kg/day	112–154 g/day	ISSN Position Stand 2022
Endurance athlete	1.2–1.6 g/kg/day	84–112 g/day	IOC 2018
Elderly (>60 years)	1.2–1.5 g/kg/day	84–105 g/day	PROT-AGE 2013; ESPEN
Pregnancy	1.1 g/kg + 25g extra	~103g/day (70kg)	ICMR; WHO

Population / Goal	Protein Intake	Total for 70kg Person	Source
Lactation (first 6 months)	1.1 g/kg + 18–20g extra	~98g/day	ICMR; WHO
Obesity / weight management	1.2–1.5 g/kg IBW/day	Based on ideal body wt	AACE 2016
CKD (pre-dialysis)	0.6–0.8 g/kg/day	42–56 g/day	KDIGO 2020
CKD (on dialysis)	1.2–1.4 g/kg/day	84–98 g/day	KDIGO 2020
Type 2 Diabetes	1.0–1.5 g/kg/day	70–105 g/day	ADA 2024
PCOS	1.2–1.8 g/kg/day	84–126 g/day	Endocrine Society

3. Per-Meal Protein Distribution

Muscle protein synthesis (MPS) is maximized at **20–40g protein per meal**, depending on body size and training status. The **leucine threshold** (2–3g leucine per meal) is the key trigger. Spreading protein across 3–4 meals is more effective for MPS than consuming large amounts in 1–2 meals.

Goal	Meals/Day	Protein/Meal	Example
General health (70kg adult)	3 meals + 1 snack	20–25g/meal	70g / 3 meals = ~23g each
Muscle building (80kg, trained)	4–5 meals	30–40g/meal	160g / 4 = 40g/meal
Weight loss (65kg)	3 meals + 2 snacks	25–30g main; 10–15g snack	~90g total
Elderly (60 years+, 70kg)	3–4 meals	30–35g/meal (higher leucine needed)	~110g total; prioritize breakfast protein

4. Protein Quality — Complete vs. Incomplete

Concept	Definition	Examples
Complete protein	Contains all 9 essential amino acids (EAAs) in adequate amounts	Eggs, meat, fish, dairy, soy, quinoa
Incomplete protein	Missing or low in one or more EAAs	Most plant foods: lentils (low methionine), wheat (low lysine), rice (low lysine)
Complementary proteins	Two or more incomplete proteins combined = complete profile	Rice + dal; roti + rajma; bread + peanut butter
PDCAAS (digestibility score)	Protein Digestibility-Corrected Amino Acid Score (0–1.0). Higher = better quality	Egg: 1.0 | Milk: 1.0 | Soy: 0.99 | Lentil: 0.52 | Wheat: 0.42
DIAAS (modern standard)	Digestible Indispensable Amino Acid Score; more accurate than PDCAAS	Egg: 1.13 | Milk: 1.18 | Soy protein: 0.97 | Lentils: 0.58–0.70

5. Protein Sources — Vegetarian & Non-Vegetarian

5A. Vegetarian Protein Sources

Food	Serving	Protein (g)	Quality Notes
Whole egg	1 large (50g)	6–7g	PDCAAS 1.0 — gold standard; all EAAs
Paneer (cottage cheese)	100g	18g	Good calcium + protein; high sat fat
Greek yogurt (plain)	150g (1 cup)	15g	Probiotic bonus; choose unsweetened
Cow's milk	250ml	8–9g	Good bioavailability
Tofu (firm)	100g	8–10g	PDCAAS 0.99; soy complete protein
Moong dal (cooked)	1 cup (200g)	14g	Easy digestibility; sprouting boosts nutrition
Masoor dal (cooked)	1 cup (200g)	18g	Iron-rich; low GI
Rajma (cooked)	1 cup (200g)	15g	High fiber + protein
Chickpeas (chana)	1 cup (200g)	15g	MUFA, fiber, potassium
Edamame (cooked)	1 cup (160g)	17g	Complete soy protein
Quinoa (cooked)	1 cup (185g)	8g	Complete protein grain; gluten-free
Peanut butter	2 tbsp (32g)	8g	High calorie; also healthy fat
Almonds	30g (23 nuts)	6g	Fiber, Vit E bonus; moderate protein
Hemp seeds	3 tbsp (30g)	10g	Complete protein; omega-3 (ALA)

5B. Non-Vegetarian Protein Sources

Food	Serving	Protein (g)	Quality Notes
Chicken breast (grilled)	100g	27–31g	PDCAAS 0.92; low fat, versatile
Egg whites	3 large (100g)	11g	Zero fat; pure albumin protein
Salmon (cooked)	100g	25g	Omega-3 EPA/DHA bonus
Tuna (canned in water)	100g	25–30g	Low cost, high protein, low fat
Mackerel / Bangda	100g	20g	Omega-3; affordable; Indian coastal staple
Lean beef	100g	26g	B12, iron, zinc; limit to 1–2×/week
Turkey breast	100g	29g	Very lean; high BCAA content
Shrimp/prawns (cooked)	100g	20g	Low fat; iodine bonus
Cottage cheese (low-fat)	100g	11g	Slow-digesting casein; good before sleep

6. Practical Examples — Protein Calculation

Scenario	Body Weight	Target Intake	How to Achieve
Sedentary woman, weight maintenance	60 kg	60–72g/day (1.0–1.2g/kg)	2 cups dal (28g) + 1 cup curd (9g) + 2 eggs (13g) + nuts (6g) = ~56g + roti/rice protein

Scenario	Body Weight	Target Intake	How to Achieve
Active man, muscle gain	80 kg	140–176g/day (1.75–2.2g/kg)	Chicken 200g (56g) + 2 eggs (13g) + paneer 100g (18g) + 2 cups dal (28g) + 2 cups milk (16g) = ~131g + supplementation if needed
Elderly woman	55 kg	66–83g/day (1.2–1.5g/kg)	Emphasize breakfast protein: egg omelette (13g) + curd bowl (9g); lunch dal 1.5 cups (21g); dinner tofu/paneer (15g) + dal (14g)
PCOS patient	72 kg	86–130g/day (1.2–1.8g/kg)	Greek yogurt (15g) + eggs (13g) + chicken 150g (40g) + rajma cup (15g) + milk (9g) = ~92g
Diabetic patient (T2DM)	70 kg	70–105g/day (1.0–1.5g/kg)	Emphasize plant protein: 2 cups mixed dal (30g) + paneer 50g (9g) + 2 eggs (13g) + curd (9g) = ~61g + cereal protein

7. Common Myths & Scientific Clarifications

■ **MYTH: High protein damages kidneys in healthy people.**

■ ***FACT: Extensive meta-analyses (Kalantar-Zadeh 2017; Lowery 2016) confirm that high protein intake (up to 2.5 g/kg/day) does NOT damage kidneys in healthy individuals. Restriction is ONLY needed in pre-existing chronic kidney disease (CKD Stage 3+).***

■ **MYTH: You can only absorb 20–30g protein per meal.**

■ ***FACT: There is no 'absorption cap' — the body absorbs all protein, but muscle protein synthesis (MPS) is maximized at 20–40g per meal. Excess beyond MPS threshold is oxidized for energy or used for other functions. Spreading protein evenly is OPTIMAL but large amounts are still digested.***

■ **MYTH: Plant protein is inferior and cannot build muscle.**

■ ***FACT: Plant proteins can fully support muscle growth when: (a) total intake is adequate, (b) complementary proteins are combined (rice+dal), and (c) leucine-rich sources like soy, edamame, or a varied diet are included. Soy protein has DIAAS >0.97.***

■ **MYTH: Protein powders are necessary.**

■ ***FACT: Whole food sources are always preferred. Supplements are only needed when whole food intake is consistently inadequate (e.g., athletes >1.8g/kg, post-surgery recovery). Whey, casein, and pea protein are safe, evidence-based options when whole foods are insufficient.***

■ **MYTH: Excess protein causes obesity.**

■ ***FACT: Protein has the highest satiety value and the highest thermic effect (~25–30% of protein calories burned in digestion). High-protein diets are consistently associated with LOWER total caloric intake and BETTER weight management outcomes in RCTs.***

References: ISSN Position Stand on Protein 2022 • ICMR-NIN RDA 2020 • PDCAAS/DIAAS FAO 2013 • Stokes et al. Nutr Rev 2018 • Morton et al. BJSM 2018